Yubo Cai

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yubocai-poly | Google Scholar | Personal Webpage | MIT CCSE Profile

Research Interests

My research bridges theory and practice across **optimization**, **learning theory**, **compositional and game-theoretic decision-making**, and **systems and control**. I study how optimization, learning, and game theory can exploit the compositional structure of large-scale systems built from heterogeneous, interacting subsystems to improve performance, robustness, scalability, and coordination. I am particularly interested in *multi-agent reinforcement learning* and distributed decision-making in interconnected systems, including transportation networks.

Education

Massachusetts Institute of Technology (MIT) Sep. 2025 – May. 2029
PhD in Computational Science & System Engineering - GPA: 5.0/5.0 Cambridge, MA, USA

Department: Center for Computational Science and Engineering & Civil and Environmental Engineering
Laboratory: MIT Laboratory for Information and Decision Systems (LIDS)
Supervisor: Prof. Gioele Zardini
Honors: SPEEDWELL Foundation Transportation Fund Fellowship, MathWorks Research Fellowship

University of Oxford, Mathematical Institute Oct. 2024 – Jul. 2025
MSc in Mathematics and Statistics (OMMS – Part C) – Graduated with Honors Oxford, United Kingdom

Direction: Statistics, Optimization, Machine Learning
Dissertation: Adaptive Levenberg-Marquardt Third-Order Newton's Method (ALMTON)
Supervisor: Prof. Coralia Cartis
Courses: Combinatorics, Probability and Statistics for Network Analysis, Theory of Deep Learning, Combinatorial Optimization, Continuous Optimization, Statistical Machine Learning
Honors: China Oxford Scholarship Award 2024–2025 (COSF 2024)

École Polytechnique (X) Aug. 2021 – Jul. 2024
BSc in Applied Mathematics and Computer Science – Graduated with Honors Paris, France

Dissertation: Feature-Incremental Machine Learning via Robust Optimization Approach
Supervisor: Prof. Gleb Pogudin, Prof. Vassilis Digalakis Jr, Prof. Michael Lingzhi Li

Research and Industrial Experience

INSEAD × BNP Paribas Jun. 2024 – May 2025
Research Associate (Predoctoral) Fontainebleau, France

Research Project: Optimal Experimentation for Learning Personalized Policies Across Locations
Algorithm Development & Applications: Developed and implemented the full **transfer-learning algorithm**, sharing information across BNP Paribas' global business units while adapting decisions to heterogeneous local data. The algorithm was **deployed at BNP Paribas** for **personalized pricing** and **risk monitoring and management**, delivering promising performance in real-world business applications.
Supervisors: Georgina Hall and Spyros Zoumpoulis (Decision Sciences)

HEC Paris × Harvard Business School Dec. 2023 – Jun. 2024
Bachelor Thesis Jouy-en-Josas, Saclay, France

Research Topics: Feature-Incremental Machine Learning via Robust Optimization Approach
Applications & Impact: The research was applied in a deployed system at **Massachusetts General Hospital (MGH)**, supporting **AI-assisted disease diagnosis and preliminary diagnostic assessment**.
Supervisor: Vassilis Digalakis Jr (HEC Paris ISOM), Michael Lingzhi Li (Harvard Business School)
Department: Information Systems and Operations Management

Center for Research in Economics and Statistics (CREST) / LIX Oct. 2023 – Jan. 2024
Project Research Intern Palaiseau, France

Research Project: BATO-MOUCHE: Urban amenities and public transport, in collaboration with **INSEE**. [GitHub]

Research Outcomes: Delivered **data-driven reports on socioeconomic inequality** to INSEE, translating experimental findings into **recommendations for more equitable access** to urban amenities and public transport.

Supervisor: Sarah J. Berkemer (LIX Aimbio Team)

HEC Montréal / Université de Montréal Jun. 2023 – Oct. 2023
Visiting Researcher (PhD Level) Montréal, Canada

Research Topics: 1. Robust Optimization of Fixed-Job Scheduling Problem. 2. Mathematical Programming Methods for Semantic Web and Logical Inference

Supervisor: Maryam Daryalal

Department: Department of Decision Sciences and Operation Research

Award: Mitacs Globalink Research Scholarship

CNRS and LIX Dec. 2022 – Aug. 2023
Research Intern then Summer Intern Palaiseau, France

Research Topics: Dissipative quadratizations of polynomial ODE systems [Paper] – published at TACAS 2024

Software: DQbee, a Python package for computing the dissipative quadratization of polynomial ODE systems

Supervisor: Gleb Pogudin

Team: MAX – Algebraic Modeling and Symbolic Computation (Joint Team with CNRS)

Publications, Preprints, and Working Papers

C = Conference P = Preprint W = Working Paper

[W.4] **Yubo Cai**, Vassilis Digalakis Jr., and Michael Lingzhi Li (2026). **Robust Learning over Streaming Feature Spaces**. *Working paper; in preparation for submission to Operations Research (OR).*

[W.3] **Yubo Cai** and Gioele Zardini (2026). **Prediction Horizons of Fixed Affine Models near a Critical Learning Rate**. *Working paper; in preparation for submission to the Learning for Dynamics and Control Conference (L4DC 2027).*

[W.2] **Yubo Cai**, Jiarui Li, and Gioele Zardini (2026). **Exact Sparsification of Safety Constraints for Multi-Robot Navigation**. *Working paper; in preparation for submission to IEEE Transactions on Robotics (T-RO).*

[W.1] **Yubo Cai** and Gioele Zardini (2026). **Exact Stability Verification of Neural Dynamics via Activation Constraint Selection**. *submitted to the International Conference on Learning Representations (ICLR 2027).*

[P.4] Alan Bonomi, Antonio Terpin, **Yubo Cai**, Yujun Huang, Meshal Alharbi, and Gioele Zardini (2026). **Compositional Sensitivity Analysis for Multiobjective System Co-Design**. *Preprint; submitted to IEEE Transactions on Automatic Control (TAC).* A conference version has been submitted to the *2027 American Control Conference (ACC 2027).* https://arxiv.org/pdf/2610.18758

[P.3] **Yubo Cai** and Gioele Zardini (2026). **Optimizing Lyapunov Certificates via Stability-Preserving Quadratization for Polynomial Systems**. *Preprint; submitted to SIAM Journal on Control and Optimization (SICON).* A conference version has been submitted to the *2027 American Control Conference (ACC 2027).* arXiv: 2609.18871.

[P.2] **Yubo Cai**, Yujun Huang, Meshal Alharbi, Gioele Zardini (2026). **Scalable Co-Design via Linear Design Problems: Compositional Theory and Algorithms**. *Preprint; major revision at IEEE Transactions on Automatic Control (TAC)*; a preliminary version was accepted to the 2026 IEEE Conference on Decision and Control (CDC 2026), see **[C.2]**. https://arxiv.org/abs/2603.29083

[P.1] **Yubo Cai**, Wenqi Zhu, Coralia Cartis, and Gioele Zardini (2026). **A Globally Convergent Third-Order Newton Method via Unified Semidefinite Programming Subproblems**. *Preprint; under review at Journal of Optimization Theory and Applications (JOTA).* https://arxiv.org/abs/2603.09682

[C.2] **Yubo Cai**, Yujun Huang, Meshal Alharbi, Gioele Zardini (2026). **Linear System Co-Design via Lifted Polyhedral Query Solving**. In *2026 IEEE Conference on Decision and Control (CDC 2026)* (accepted). Preliminary conference version of **[P.2]**.

[C.1] **Yubo Cai** and Gleb Pogudin (2024). **Dissipative Quadratizations of Polynomial ODE Systems**. In *Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2024)*, pp. 323–342. Springer Nature Switzerland. DOI: 10.1007/978-3-031-57249-4_16

Teaching Experience

HEC Montréal 2023 Summer
Teaching Assistant Editor Montréal, Canada

Course: MATH 20604A – Linear Optimisation Models

Content: Linear Programming, Integer Programming, Simplex, Column Generation, Cutting Plane

Presentations and Talks

2026 Northeast Systems and Control Workshop (NESCW): *Quadratization-induced pullback SOS for scalable Lyapunov synthesis*, May 2026, Princeton, NJ

2026 INFORMS Optimization Society: *A Globally Convergent Third-Order Newton Method via Unified Semidefinite Programming Subproblems*, Mar 2026, Atlanta, GA

2026 MIT CEE Research Day: *Linear Co-Design Problems: Composition Theory, Solving Algorithms, and Approximation*, Jan 2026, Cambridge, MA

2026 MIT LIDS Student Conference: *A Globally Convergent Third-Order Newton Method via Unified Semidefinite Programming Subproblems*, Jan 2026, Cambridge, MA

ETAPS 2024: *Dissipative quadratizations of polynomial ODE systems*, Apr 2024, Luxembourg **[slide]**

Hi!Paris BATO-MOUCHE Big-City Analysis Seminar: *Analysis of city composition based on basic amenities including public transport*, Jan 2024, CREST, Paris-Saclay, France **[webpage]**

The MAX Computer Algebra Seminar: *Quadratization in the reachability problem for ODEs*, Jun 2023, INRIA, Palaiseau, France **[webpage]**

Projects

DQbee: A Python Library to compute the optimal inner-quadratic quadratization and dissipative quadratization of a given polynomial ODE system

BATO-MOUCHE Project: Analysis of city composition based on basic amenities including public transport

Reachability Problem of Nonlinear System: Research project on finding the optimal quadratization method which fits the reachability error bound function

Nim Game with Coq: Using the language Coq to prove the winning strategy of the Nim Game

Pricing an Option with Binomial Models: Using binomial models in the pricing of options with different time periods

Honors and Awards

[1] **IEEE CDC26 Student Travel Award** – Amount: $1500

[2] **MathWorks Research Fellowship** – Amount: $118,521

[3] **NESCW26 Travel Grant** – Amount: $200

[4] **MIT CEE Ippen Travel Grant** – Amount: $1,000 $\times 2$

[5] **SPEEDWELL Foundation Transportation Fund Fellowship** – Amount: $117,521

[6] **China Oxford Scholarship Fund 2024–25 Awards** – Amount: 5,500 GBP

[7] **INFORMS Undergraduate Scholarship** – Amount: $1,250 (Unable to attend the meeting due to schedule conflict)

[8] **Mitacs Globalink Research Scholarship** – Amount: 6,000 €

[9] **ETAPS Student Scholarship** – Amount: 1,000 €

[10] **ANR JCJC Project OCCAM Research Intern Scholarship** – Amount: 1,200 €

Previous Students

Weihan Zhang (Caltech Undergrad) - Summer 2026, **Sevinj Mahmudova** (MIT Undergrad) - Fall 2026